

SMART INDIA HACKATHON 2026



Problem Statement ID: SIH26105

Problem Statement Title: AI-Powered Continuous Cyber Risk Quantification & Investment Optimization Platform

Theme: Blockchain & Cybersecurity

PS Category: Software

Team ID: 180219

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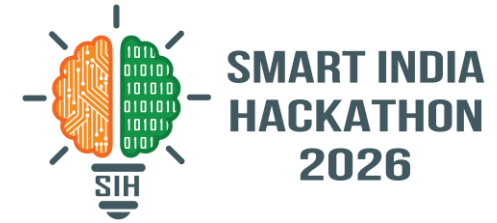
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Institute: Skyline Institute of Engineering and Technology, Greater Noida



(ID: 180219)

IDEA TITLE: AURA SENTINEL (AI-POWERED CYBER RISK QUANTIFICATION)



Real-World Issue

Enterprises rely on static, qualitative 'High/Medium/Low' heatmaps that hide actual financial exposure, leaving boards unable to justify cyber budgets against dynamic threats.

Why Important

Cyber incidents cost Indian enterprises ₹25,000+ Cr annually. 68% of CISOs struggle to defend security ROI to boards without quantified financial risk metrics.

Proposed Solution: AURA Sentinel

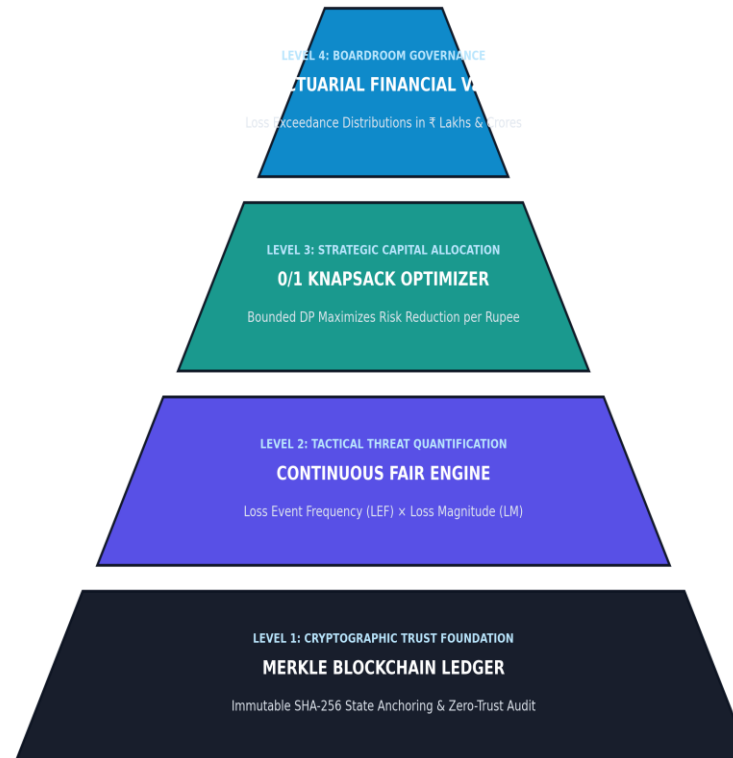
AI-powered continuous cyber risk quantification combining the Open Group FAIR standard, Monte Carlo simulation, 0/1 Knapsack optimization, and SHA-256 Merkle blockchain.

• Live SOC Console

• GitHub Repo

• REST API Docs

AURA CYBER DEFENSE MATURITY PYRAMID



RISK: Static Annual Heatmaps

Subjective High/Medium guesswork; stale within days; zero actuarial Rupee quantification.

vs

SOLUTION: Continuous FAIR Engine

Automated continuous loss calculation (Threat Frequency × Loss Magnitude in ₹).

RISK: Arbitrary Budget Allocation

Security budgets allocated via vendor hype & panic; no mathematical ROI justification.

vs

SOLUTION: 0/1 Knapsack Optimizer

Bounded dynamic programming maximizes risk reduction per Rupee invested.

RISK: Repudiable Audit Trails

Editable database logs that fail rigorous regulatory scrutiny (DPDP & CERT-In).

vs

SOLUTION: Merkle Blockchain Ledger

Cryptographically sealed SHA-256 state anchoring for zero-trust compliance.

TECHNICAL APPROACH & SYSTEM ARCHITECTURE



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TECHNOLOGIES USED

- **Backend & APIs:**
Python 3.10, FastAPI, Pydantic, Uvicorn, SQLite 3
- **Quantification Engine:**
Open Group FAIR (O-RT), NumPy, Monte Carlo (10k)
- **Optimization Logic:**
0/1 Knapsack (Dynamic Programming), Multi-Constraint
- **Blockchain & Audit:**
SHA-256 Merkle Tree, Cryptographic Hash Chain
- **Frontend & SOC UI:**
React 18, Tailwind CSS, Lucide Icons, Recharts

FEASIBILITY, VIABILITY & BUSINESS POTENTIAL

FEASIBILITY ANALYSIS

✓ Plug-and-Play Ingestion

Seamlessly connects with existing tools (Nessus, Qualys, AWS, Active Directory) via standard REST APIs without hardware changes.

✓ Sub-Second Computation

10,000 Monte Carlo runs execute in <850ms in pure NumPy; dynamic programming knapsack completes in <50ms.

✓ Modular Microservices

Dockerized, container-ready architecture allows independent scaling of analytics workers and API nodes.

✓ Agentless Deployment

Zero endpoint installation required, drastically reducing IT friction and operational overhead.

VIABILITY & GOVERNANCE

✓ Global Actuarial Standards

Built strictly on Open Group FAIR & NIST SP 800-30 Rev 1, accepted by boards and cyber insurers globally.

✓ Regulatory Compliance

Merkle audit proofs satisfy DPDP Act 2023, RBI Cyber Security Framework, and CERT-In logging directives.

✓ Boardroom Defensibility

Converts ambiguous technical CVSS scores into CFO-ready ₹ Rupee loss distributions and justified ROI.

✓ Zero Vendor Lock-In

Open JSON/CSV data schemas and open-standards compliance ensure transparent, portable risk registers.

BUSINESS POTENTIAL & ROI

✓ 70%+ Audit Cost Reduction

Automates manual third-party cyber risk assessments, saving ₹20L–₹50L per annual enterprise audit cycle.

✓ Optimized Security Spend

Dynamic knapsack allocation prevents overspending on low-impact controls, saving 25–40% of security budget.

✓ Enterprise Market Scale

Targeted at BFSI, Healthcare, Critical Infrastructure, and IT enterprises needing continuous compliance.

✓ Cyber Insurance Alignment

Empirical Value-at-Risk modeling accelerates policy underwriting and lowers corporate insurance premiums.

IMPACT AND BENEFITS

AURA SENTINEL // MULTI-DIMENSIONAL IMPACT MATRIX

OPERATIONAL ADVANTAGES

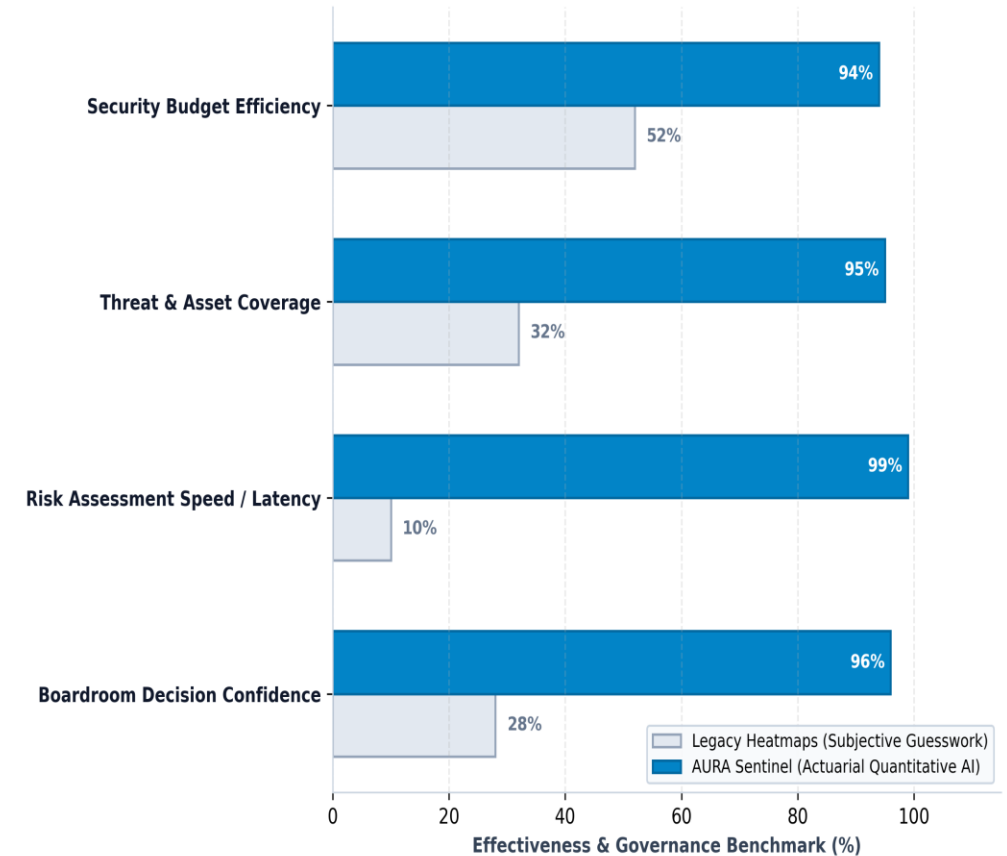
- **24/7 Continuous Visibility**
Replaces periodic static audits with real-time dynamic risk quantification.
- **Optimal Budget ROI**
Knapsack algorithm maximizes security risk reduction per Rupee invested.
- **Impact Remediation**
Fixes vulnerabilities based on financial loss risk rather than CVSS noise alone.
- **Cryptographic Audit**
Tamper-proof Merkle ledger guarantees verifiable non-repudiation.

STRATEGIC & REGULATORY PILLARS

- **Boardroom Financial Clarity**
Translates cyber exposure into Value-at-Risk (₹ Crores) for executive consensus.
- **Compliance Readiness**
Automates reporting for DPDP Act 2023, RBI Cyber Guidelines, & CERT-In.
- **Cyber Insurance Underwriting**
Provides empirical loss exceedance curves required for favorable policy rates.
- **Unified C-Suite Vocabulary**
Bridges the technical communication gap between SOC analysts and CFO/Board.

★ Paradigm Shift: From Subjective Guesswork to Actuarial Financial Precision ★

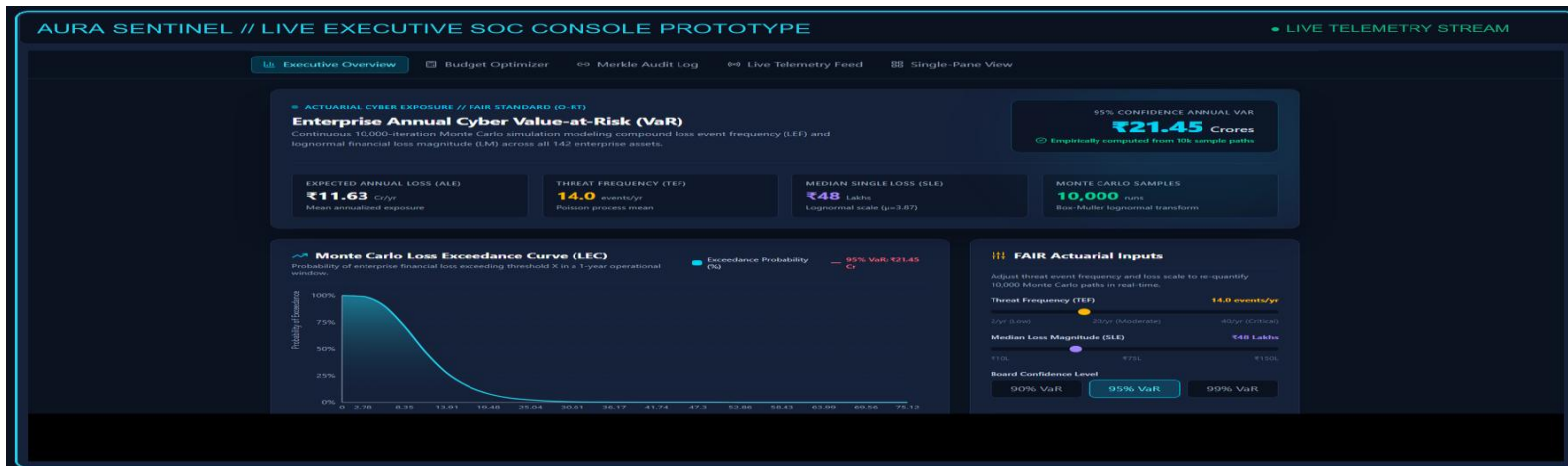
PERFORMANCE BENCHMARK: LEGACY VS AURA SENTINEL



STANDARDS, FORMAL RESEARCH & REGULATORY CITATIONS

- **The Open Group FAIR Standard:** Risk Taxonomy (O-RT) & Risk Analysis (O-RA) for quantitative information risk modeling. [opengroup.org]
- **NIST SP 800-30 Rev. 1:** Guide for Conducting Risk Assessments in Federal Information Systems. [nist.gov]
- **ISO/IEC 27005:2022:** Information security, cybersecurity and privacy protection — Managing risks. [iso.org]
- **DPDP Act 2023:** Digital Personal Data Protection Act, Ministry of Electronics & IT (MeitY), India. [meity.gov.in]
- **CERT-In Directions (2022):** Section 70B mandate for cybersecurity posture and log retention guidelines. [cert-in.org.in]
- **0/1 Knapsack Optimization:** Dynamic Programming algorithms for resource-constrained risk minimization. [ieee.org]
- **RFC 6962: Merkle Trees:** Verifiable append-only data structures for immutable security telemetry logging. [rfc-editor.org]

OFFICIAL CODE REPOSITORY



github.com/aryanbaghel756-spec/AURA-SENTINEL